In [46]:
```
### Training with fancier version ###

import torch
import torch.nn as nn
import torch.nn.functional as F

import torch.optim as optim
import matplotlib.pyplot as plt

class Net(nn.Module): ## nn.Module class is used
    def __init__(self):
        super().__init__()

        self.fc1 = nn.Linear(1, 16, bias=True)
        self.act1 = nn.Tanh()

        self.fc2 = nn.Linear(16, 16, bias=True)
        self.act2 = nn.Tanh()

        self.fc3 = nn.Linear(16, 1, bias=True)

    def forward(self, x):
        x = self.act1(self.fc1(x))
        x = self.act2(self.fc2(x))
        x = self.fc3(x)  # No output activation
        return x

net = Net()

print(net)
print(list(net.parameters())) # parameters are randomized


#def criterion(out, label):
#    return (label - out)**2
criterion = nn.MSELoss()


# optimizer = optim.SGD(net.parameters(), lr=0.1)
optimizer = optim.Adam(net.parameters(), lr=0.01)

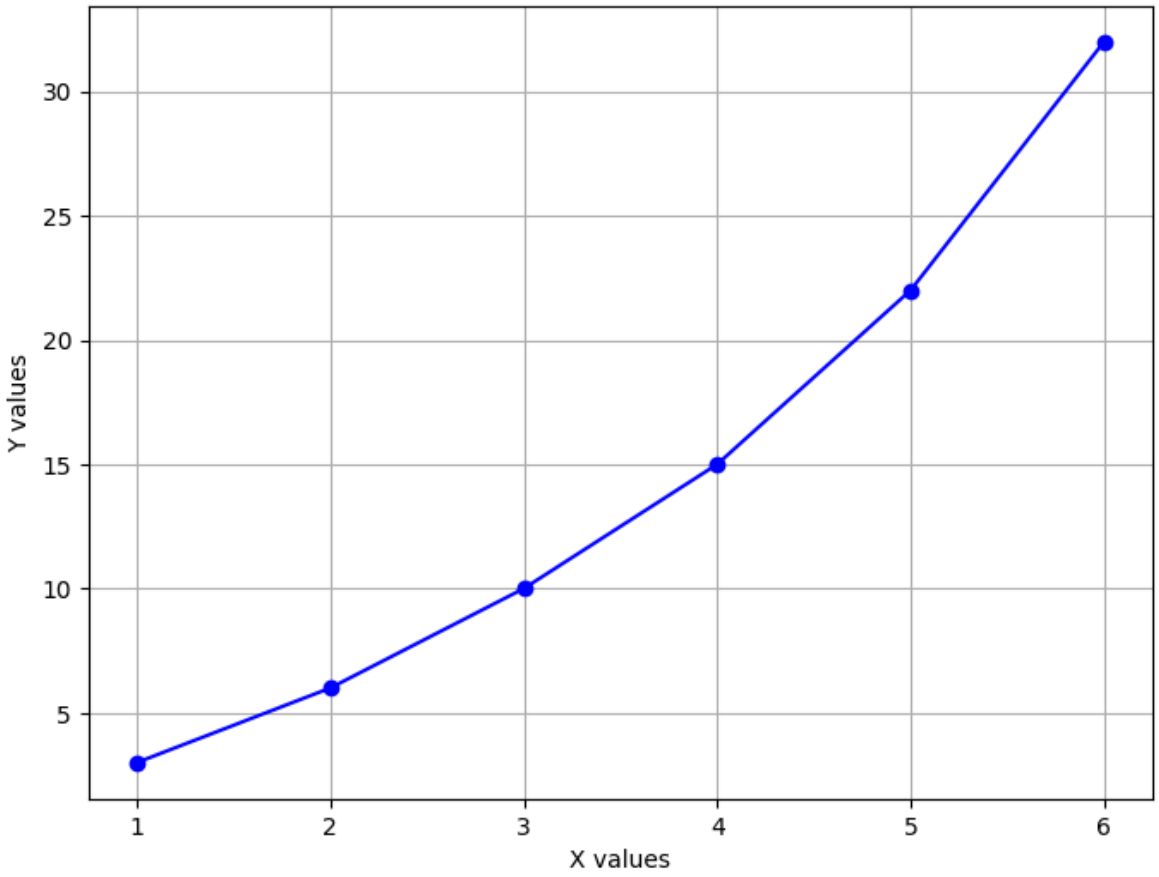

data = [(1.0,3.0), (2.0,6.0), (3.0,10.0), (4.0,15.0), (5.0,22.0), (6.0,32.0)]

# Split data into x and y
x_values, y_values = zip(*data)

# Plotting
plt.figure(figsize=(8, 6))
plt.plot(x_values, y_values, marker='o', linestyle='-', color='b')
plt.xlabel('X values')
plt.ylabel('Y values')
plt.grid(True)
plt.show()
```

```
Net(
  (fc1): Linear(in_features=1, out_features=16, bias=True)
  (act1): Tanh()
  (fc2): Linear(in_features=16, out_features=16, bias=True)
  (act2): Tanh()
  (fc3): Linear(in_features=16, out_features=1, bias=True)
)
[Parameter containing:
tensor([[ 0.3066],
        [-0.9229],
        [-0.8026],
        [ 0.1302],
        [ 0.5766],
        [ 0.2646],
        [ 0.2618],
        [ 0.2404],
        [-0.2466],
        [-0.2804],
        [ 0.1131],
        [ 0.0394],
        [ 0.9090],
        [ 0.5970],
        [ 0.6365],
        [-0.6608]], requires_grad=True), Parameter containing:
tensor([ 0.1325,  0.8033, -0.5506, -0.3546,  0.5854,  0.9352, -0.8683, -0.1927,
         0.8159, -0.0212,  0.1570, -0.6998,  0.8107, -0.0864, -0.9675,  0.2983],
       requires_grad=True), Parameter containing:
tensor([[ 1.2486e-01, -1.1504e-01,  2.1264e-01, -2.8161e-02, -1.1712e-05,
         -2.1211e-01,  1.1329e-01,  7.5698e-02, -1.9750e-01, -4.9676e-02,
         -1.4615e-01,  1.0684e-01, -2.1172e-01,  3.6822e-02,  1.8927e-01,
          8.5638e-02],
        [-1.0363e-01,  8.1149e-02,  3.8339e-02, -1.8632e-01, -8.1954e-02,
         -1.4963e-01,  9.6096e-02,  1.5700e-01, -5.6401e-02,  1.9558e-01,
         -6.8962e-02,  5.7712e-03,  1.3654e-01,  9.6084e-02,  1.4118e-02,
          2.3547e-01],
        [ 2.3134e-01, -5.4165e-02,  1.2739e-01,  5.7898e-02,  6.3178e-02,
         -2.3039e-01, -3.4426e-03,  1.9623e-01,  1.5961e-02, -1.3035e-01,
          1.7061e-01,  4.5013e-02,  2.2824e-01, -1.7025e-01, -4.1148e-03,
          1.3029e-01],
        [-1.9331e-01, -1.6592e-02, -7.2121e-02, -1.4953e-01, -2.4795e-01,
         -8.3776e-02, -2.1978e-01, -1.3275e-01, -1.3984e-01,  2.4766e-01,
          9.4625e-02, -2.1244e-01, -6.7243e-02,  1.2447e-02,  1.8624e-01,
          5.9858e-03],
        [-5.5712e-02, -1.1187e-01,  1.8588e-01,  1.6858e-01,  2.5333e-02,
         -1.7684e-01,  4.2009e-02,  1.2596e-01,  3.4698e-02,  1.9484e-02,
         -1.6755e-01, -6.4004e-02, -4.4138e-02,  1.1440e-01,  1.6970e-01,
         -2.4199e-02],
        [ 1.9406e-01, -1.4151e-01, -1.8894e-01, -1.0455e-01,  8.5770e-02,
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          1.7937e-01],
        [ 1.8826e-01,  2.3717e-01, -1.7893e-01, -1.6329e-01, -1.7726e-01,
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          9.5414e-02, -1.6232e-01, -2.4632e-01, -2.4110e-01,  1.3283e-01,
          1.6533e-01],
        [-3.6229e-02, -8.2780e-02, -5.3247e-02,  1.9408e-01,  3.1957e-02,
```

```
          -1.4069e-01,  1.2640e-01, -1.2206e-01,  1.2637e-01, -1.5519e-01,
           1.9787e-01,  3.8733e-03, -2.1035e-02,  1.2760e-01,  1.5308e-01,
          -8.8278e-02],
         [ 8.8998e-02, -2.3776e-01, -1.7787e-01,  1.6976e-01,  1.1759e-01,
          -2.5404e-02,  2.1440e-01,  5.4355e-02,  7.0001e-02,  1.9404e-01,
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           3.2773e-02],
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          -2.3574e-01,  4.4308e-02, -1.6282e-01, -9.5886e-02,  8.2362e-02,
           9.9168e-02],
         [-8.9462e-02,  1.5308e-01,  1.9630e-02, -2.4289e-01, -2.7170e-02,
           2.1260e-01,  1.1191e-01,  1.6689e-01,  1.5487e-01,  4.5005e-02,
           1.9168e-01, -1.6630e-01,  8.7164e-02, -1.4941e-01, -1.5115e-01,
           1.8989e-01],
         [-2.3679e-01,  2.4536e-04,  2.2565e-01,  1.5572e-02, -9.2580e-02,
           5.0499e-02,  1.4292e-01, -2.2007e-02,  1.9103e-01,  1.8569e-01,
          -2.0925e-01, -3.5329e-02, -1.8853e-02,  1.4537e-01,  1.5497e-02,
           1.5717e-01],
         [-2.1134e-01, -2.0270e-01, -1.1706e-01, -6.0780e-02, -2.3009e-01,
          -1.8331e-01,  2.4494e-01, -2.1921e-01, -1.6089e-01,  5.7390e-03,
           1.8717e-02,  1.2163e-01,  1.5871e-01,  1.3625e-01,  1.7880e-01,
           2.0462e-01],
         [ 7.1357e-02, -2.3653e-01, -4.4281e-02, -2.3188e-01, -1.4207e-01,
          -8.7587e-02,  1.5591e-01, -4.0104e-02, -2.2007e-01, -2.2390e-01,
           6.4223e-02,  6.0044e-03, -1.6274e-01,  2.0642e-03, -1.6970e-01,
          -1.5530e-01],
         [ 4.2941e-02,  1.6141e-01, -1.7654e-01,  1.6661e-01, -9.4309e-02,
          -1.6683e-01, -2.1623e-01,  2.1615e-01, -3.4449e-02, -1.6880e-02,
          -2.4114e-01, -7.5681e-03, -1.7238e-01, -1.9718e-01,  2.7711e-02,
          -1.2711e-01]], requires_grad=True), Parameter containing:
tensor([-0.1667, -0.2280,  0.1368, -0.0492, -0.2348, -0.0906,  0.0269,  0.0022,
        -0.1578,  0.1463,  0.0063, -0.2209, -0.1626, -0.2366,  0.2475,  0.1313],
       requires_grad=True), Parameter containing:
tensor([[ 0.1427, -0.1889, -0.1234, -0.0654, -0.1962,  0.1276,  0.1773,  0.2232,
         -0.0556,  0.2215,  0.0580, -0.1891,  0.1508,  0.0231,  0.1620, -0.061
2]],
       requires_grad=True), Parameter containing:
tensor([0.1840], requires_grad=True)]
```




In [47]:
```
for epoch in range(1000):
    for i, current_data in enumerate(data):
        X, Y = current_data
        X, Y = torch.FloatTensor([X]), torch.FloatTensor([Y])
        optimizer.zero_grad()
        outputs = net(X)
        loss = criterion(outputs, Y)
        loss.backward()
        optimizer.step()    ## This line is equivalent to "W = W - lr* W.grad"
    print("Epoch {} - loss: {}".format(epoch, loss))


### Test the trained network ###
for i, current_data in enumerate(data):
    X, Y = current_data
    X, Y = torch.FloatTensor([X]), torch.FloatTensor([Y])
    out = net(torch.FloatTensor(X))
    print("when x = {}, y = {}".format(X, out))
```

```
Epoch 0 - loss: 928.57861328125
Epoch 1 - loss: 858.5050659179688
Epoch 2 - loss: 802.590087890625
Epoch 3 - loss: 754.3414306640625
Epoch 4 - loss: 709.4274291992188
Epoch 5 - loss: 667.6375732421875
Epoch 6 - loss: 629.424072265625
Epoch 7 - loss: 594.7835083007812
Epoch 8 - loss: 563.53173828125
Epoch 9 - loss: 535.4067993164062
Epoch 10 - loss: 510.1085510253906
Epoch 11 - loss: 487.3191223144531
Epoch 12 - loss: 466.70892333984375
Epoch 13 - loss: 447.9522705078125
Epoch 14 - loss: 430.67742919921875
Epoch 15 - loss: 413.66021728515625
Epoch 16 - loss: 394.3923645019531
Epoch 17 - loss: 374.682373046875
Epoch 18 - loss: 356.69561767578125
Epoch 19 - loss: 340.4415283203125
Epoch 20 - loss: 325.405029296875
Epoch 21 - loss: 310.7611083984375
Epoch 22 - loss: 296.2897644042969
Epoch 23 - loss: 282.58563232421875
Epoch 24 - loss: 269.70654296875
Epoch 25 - loss: 257.29998779296875
Epoch 26 - loss: 245.40037536621094
Epoch 27 - loss: 234.19195556640625
Epoch 28 - loss: 223.5659942626953
Epoch 29 - loss: 213.4186248779297
Epoch 30 - loss: 203.80245971679688
Epoch 31 - loss: 194.69949340820312
Epoch 32 - loss: 186.04119873046875
Epoch 33 - loss: 177.82168579101562
Epoch 34 - loss: 170.01943969726562
Epoch 35 - loss: 162.576171875
Epoch 36 - loss: 155.46539306640625
Epoch 37 - loss: 148.66656494140625
Epoch 38 - loss: 142.15110778808594
Epoch 39 - loss: 135.91409301757812
Epoch 40 - loss: 129.94515991210938
Epoch 41 - loss: 124.23577880859375
Epoch 42 - loss: 118.78050994873047
Epoch 43 - loss: 113.56961059570312
Epoch 44 - loss: 108.59790802001953
Epoch 45 - loss: 103.85639953613281
Epoch 46 - loss: 99.3379135131836
Epoch 47 - loss: 95.03252410888672
Epoch 48 - loss: 90.932373046875
Epoch 49 - loss: 87.02804565429688
Epoch 50 - loss: 83.3121337890625
Epoch 51 - loss: 79.77536010742188
Epoch 52 - loss: 76.41130065917969
Epoch 53 - loss: 73.21035766601562
Epoch 54 - loss: 70.16889953613281
Epoch 55 - loss: 67.27586364746094
Epoch 56 - loss: 64.5313720703125
Epoch 57 - loss: 61.92073440551758
Epoch 58 - loss: 59.445526123046875
Epoch 59 - loss: 57.08140182495117
```

```
Epoch 60 - loss: 54.8227653503418
Epoch 61 - loss: 52.63766860961914
Epoch 62 - loss: 50.52572250366211
Epoch 63 - loss: 48.46856689453125
Epoch 64 - loss: 46.481727600097656
Epoch 65 - loss: 44.549095153808594
Epoch 66 - loss: 42.68949890136719
Epoch 67 - loss: 40.875247955322266
Epoch 68 - loss: 39.129669189453125
Epoch 69 - loss: 37.430904388427734
Epoch 70 - loss: 35.733642578125
Epoch 71 - loss: 34.1259880065918
Epoch 72 - loss: 32.32182312011719
Epoch 73 - loss: 29.768949508666992
Epoch 74 - loss: 31.934091567993164
Epoch 75 - loss: 26.668506622314453
Epoch 76 - loss: 29.49022674560547
Epoch 77 - loss: 27.19465446472168
Epoch 78 - loss: 24.3135929107666
Epoch 79 - loss: 25.869224548339844
Epoch 80 - loss: 24.193395614624023
Epoch 81 - loss: 22.128480911254883
Epoch 82 - loss: 23.853500366210938
Epoch 83 - loss: 19.97599983215332
Epoch 84 - loss: 21.63085174560547
Epoch 85 - loss: 18.768627166748047
Epoch 86 - loss: 19.5647029876709
Epoch 87 - loss: 17.070018768310547
Epoch 88 - loss: 17.9758358001709
Epoch 89 - loss: 15.280224800109863
Epoch 90 - loss: 16.9137020111084
Epoch 91 - loss: 13.76974105834961
Epoch 92 - loss: 15.493258476257324
Epoch 93 - loss: 12.557453155517578
Epoch 94 - loss: 13.759282112121582
Epoch 95 - loss: 11.427372932434082
Epoch 96 - loss: 12.183520317077637
Epoch 97 - loss: 10.25853157043457
Epoch 98 - loss: 11.076251029968262
Epoch 99 - loss: 9.10930061340332
Epoch 100 - loss: 10.421207427978516
Epoch 101 - loss: 8.091416358947754
Epoch 102 - loss: 9.61473274230957
Epoch 103 - loss: 7.284021377563477
Epoch 104 - loss: 8.250496864318848
Epoch 105 - loss: 6.590112686157227
Epoch 106 - loss: 7.0417585372924805
Epoch 107 - loss: 5.891495704650879
Epoch 108 - loss: 6.207215309143066
Epoch 109 - loss: 5.189161777496338
Epoch 110 - loss: 5.728775501251221
Epoch 111 - loss: 4.527492046356201
Epoch 112 - loss: 5.587884426116943
Epoch 113 - loss: 3.9695253372192383
Epoch 114 - loss: 5.00447416305542
Epoch 115 - loss: 3.5579588413238525
Epoch 116 - loss: 3.999481201171875
Epoch 117 - loss: 3.2084133625030518
Epoch 118 - loss: 3.3144404888153076
Epoch 119 - loss: 2.841724157333374
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Epoch 120 - loss: 2.8693718910217285
Epoch 121 - loss: 2.4682719707489014
Epoch 122 - loss: 2.583547592163086
Epoch 123 - loss: 2.1145057678222656
Epoch 124 - loss: 2.564037322998047
Epoch 125 - loss: 1.7990319728851318
Epoch 126 - loss: 2.900362014770508
Epoch 127 - loss: 1.5723669528961182
Epoch 128 - loss: 2.0410242080688477
Epoch 129 - loss: 1.4532569646835327
Epoch 130 - loss: 1.5029586553573608
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Epoch 134 - loss: 1.0293720960617065
Epoch 135 - loss: 1.0324784517288208
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Epoch 137 - loss: 0.9359394907951355
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Epoch 140 - loss: 0.6072620153427124
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Epoch 152 - loss: 0.292380154132843
Epoch 153 - loss: 0.24079936742782593
Epoch 154 - loss: 0.25894200801849365
Epoch 155 - loss: 0.19782786071300507
Epoch 156 - loss: 0.2388136237859726
Epoch 157 - loss: 0.15984714031219482
Epoch 158 - loss: 0.2496853917837143
Epoch 159 - loss: 0.12886540591716766
Epoch 160 - loss: 0.3058886229991913
Epoch 161 - loss: 0.10787765681743622
Epoch 162 - loss: 0.27079537510871887
Epoch 163 - loss: 0.0976252555847168
Epoch 164 - loss: 0.1747090369462967
Epoch 165 - loss: 0.08787448704242706
Epoch 166 - loss: 0.12000569701194763
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Epoch 168 - loss: 0.08872123807668686
Epoch 169 - loss: 0.06858758628368378
Epoch 170 - loss: 0.07162050902843475
Epoch 171 - loss: 0.05747910588979721
Epoch 172 - loss: 0.06041577458381653
Epoch 173 - loss: 0.04771064221858978
Epoch 174 - loss: 0.05096477270126343
Epoch 175 - loss: 0.03886633366346359
Epoch 176 - loss: 0.04351199045777321
Epoch 177 - loss: 0.030962111428380013
Epoch 178 - loss: 0.038331225514411926
Epoch 179 - loss: 0.024140650406479836
```

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Epoch 180 - loss: 0.03571924567222595
Epoch 181 - loss: 0.018258551135659218
Epoch 182 - loss: 0.03716174513101578
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Epoch 184 - loss: 0.04583876207470894
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Epoch 215 - loss: 2.4028486222960055e-05
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Epoch 217 - loss: 9.799296094570309e-07
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Epoch 219 - loss: 6.338814273476601e-06
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Epoch 221 - loss: 3.1917603337205946e-05
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Epoch 227 - loss: 0.00014623074093833566
Epoch 228 - loss: 0.007376077584922314
Epoch 229 - loss: 0.00018359720706939697
Epoch 230 - loss: 0.006320585031062365
Epoch 231 - loss: 0.00022372465173248202
Epoch 232 - loss: 0.005620537791401148
Epoch 233 - loss: 0.0002690649125725031
Epoch 234 - loss: 0.005207995418459177
Epoch 235 - loss: 0.00032145180739462376
Epoch 236 - loss: 0.005029509775340557
Epoch 237 - loss: 0.00038206600584089756
Epoch 238 - loss: 0.00504792295396328
Epoch 239 - loss: 0.0004508235142566263
```

```
Epoch 240 - loss: 0.005235009361058474
Epoch 241 - loss: 0.0005278929602354765
Epoch 242 - loss: 0.0055657620541751385
Epoch 243 - loss: 0.0006125504150986671
Epoch 244 - loss: 0.006006746552884579
Epoch 245 - loss: 0.0007037032046355307
Epoch 246 - loss: 0.0065081678330898285
Epoch 247 - loss: 0.0008000966627150774
Epoch 248 - loss: 0.007000293117016554
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Epoch 659 - loss: 3.637978807091713e-12
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Epoch 699 - loss: 0.0
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Epoch 702 - loss: 0.0
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Epoch 704 - loss: 0.0
Epoch 705 - loss: 0.0
Epoch 706 - loss: 0.0
Epoch 707 - loss: 0.0
Epoch 708 - loss: 0.0
Epoch 709 - loss: 0.0
Epoch 710 - loss: 0.0
Epoch 711 - loss: 0.0
Epoch 712 - loss: 0.0
Epoch 713 - loss: 0.0
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Epoch 719 - loss: 0.0
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Epoch 720 - loss: 0.0
Epoch 721 - loss: 0.0
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Epoch 724 - loss: 0.0
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Epoch 727 - loss: 0.0
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Epoch 737 - loss: 0.0
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Epoch 750 - loss: 0.0
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Epoch 768 - loss: 0.0
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Epoch 780 - loss: 0.0
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Epoch 800 - loss: 0.0
Epoch 801 - loss: 0.0
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Epoch 840 - loss: 0.0
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Epoch 900 - loss: 0.0
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Epoch 997 - loss: 0.0
Epoch 998 - loss: 0.0
Epoch 999 - loss: 0.0
when x = tensor([1.]), y = tensor([3.0000], grad_fn=<AddBackward0>)
when x = tensor([2.]), y = tensor([6.0000], grad_fn=<AddBackward0>)
when x = tensor([3.]), y = tensor([10.0000], grad_fn=<AddBackward0>)
when x = tensor([4.]), y = tensor([15.], grad_fn=<AddBackward0>)
when x = tensor([5.]), y = tensor([22.], grad_fn=<AddBackward0>)
when x = tensor([6.]), y = tensor([32.], grad_fn=<AddBackward0>)
```

In [ ]:
```
X
```

In [ ]:
```
for epoch in range(20):
    print(epoch)
```

In [ ]:
```
W = torch.tensor([1.0], requires_grad=True)
W = W*2
label = 1.0
loss = W*5 - label
loss.backward()
W.grad
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]: